1 Proof that the Galois group of any irreducible factor of $x^n + ax + b$ is symmetric for different small values of a and b

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1.1 Summary of results

Fix $n > 0$ and $a, b \neq 0$ and let

$$f = x^n + ax + b$$

for legacy reasons define also

$$f_{n,a} = x^n + ax + 1$$

If all the conditions in one of the rows of the table below are satisfied, then the Galois group of any irreducible factor of f is the full symmetric group.

$a =$	$b =$	$n \not\equiv$	mod
± 1	± 1		
± 2	1		
± 2	± 2		
± 3	-3		
± 4	-2		
± 3	1	$-a$	18
± 4	1	4	8
5	1	$10, 40, 45, 60, 90, 95$	100
-5	1	$5, 10, 40, 55, 60, 90$	100

1.2 Notations

Fix an algebraic closure $\overline{\mathbb{Q}}$ of $\mathbb{Q}$ and for each prime p, fix an algebraic closure $\overline{\mathbb{Q}_p}$ of $\mathbb{Q}_p$ and an embedding $\overline{\mathbb{Q}} \hookrightarrow \overline{\mathbb{Q}_p}$ extending the natural embedding $\mathbb{Q} \hookrightarrow \mathbb{Q}_p$. Denote by G the absolute Galois group $\mathrm{Gal}\left(\overline{\mathbb{Q}}/\mathbb{Q}\right)$, and for each prime p, denote by $I(p) \subseteq G$ the absolute inertia group at p, i.e., the embedding into G of the inertia group $\mathrm{Gal}\left(\overline{\mathbb{Q}_p}/\mathbb{Q}_p^{\mathrm{un}}\right)$, where $\mathbb{Q}_p^{\mathrm{un}}$ is the maximal unramified extension of $\mathbb{Q}_p$ inside $\overline{\mathbb{Q}_p}$.

Fix an irreducible factor g of f (which most of the time will be f

itself) and a root $\alpha \in \overline{\mathbb{Q}}$ of g and denote $L = \mathbb{Q}(\alpha)$.

For each polynomial h and number field K, we will denote by $\mathrm{d}(h)$ and $\mathrm{d}(K)$ (or for legacy reasons also $\mathrm{disc}(K)$) the discriminant of h and of K respectively and by $\mathrm{rd}(K)$ the root discriminant of K, i.e. $\mathrm{rd}(K) = \mathrm{d}(K)^{1/[K:\mathbb{Q}]}$.

1.3 Strategy

We will first lay down a few results that will provide us with a strategy to show that for any triple n, a, b that satisfies different conditions, the Galois group of any factor of f is symmetric.

Lemma 1.1. *For any $p \nmid ab$, the action of the inertia group $I(p)$ on $\mathrm{Hom}_{\mathbb{Q}}\left(\mathbb{Q}[x]/(f), \overline{\mathbb{Q}}\right)$ has at most two non-fixed points.*

I think I saw somewhere a proof that covers this for the irreducible case, but I can't find it now. So I will try to adapt Martin's proof

Proof. If $I(p)$ acts trivially, we are done. Assume that it does not. It suffices to show that the roots of f has a single orbit under $I(p)$, and that the length of this orbit is 2. If two roots are in the same orbit under $I(p)$, then they have the same reduction modulo p. Therefore,

if we denote by $\overline{f} \in \mathbb{F}_p[x]$ the reduction of f modulo p, then it suffices to show that f has exactly one root in $\overline{\mathbb{F}_p}$ of multiplicity larger than 1 and that its multiplicity is in fact 2.

Firstly, note that by (**?**, Theorem 2)

$$\mathrm{d}(f) = (-1)^{\frac{n(n-1)}{2}} \left(n^n b^{n-1} + (1-n)^{n-1} a^n \right)$$

and hence, since $p | \mathrm{d}(f)$ and since $p \nmid ab$, it follows that p divides n iff it divides $(n-1)$ and thus it divides neither.

Next, since

$$f' = nx^{n-1} + a$$
$$f'' = n(n-1)x^{n-2}$$

and since $p \nmid n(n-1), a$ it follows that f', f'' have no shared root in $\overline{\mathbb{F}_p}$ and hence f has no triple roots in $\overline{\mathbb{F}_p}$.

Finally, note that any double root of f is also a solution of the linear equation

$$g := nf - xf' = nx^n + nax + nb - nx^n - ax = a(n-1)x + bn$$

and hence since p does not divide $a(n-1)$ there is exactly one double root of f in $\overline{\mathbb{F}_p}$. $\square$

Corollary 1.2. *Let g be an irreducible factor of f, then for any $p \nmid ab$, the action of the inertia group $I(p)$ on $\mathrm{Hom}_{\mathbb{Q}}\left(\mathbb{Q}[x]/(g), \overline{\mathbb{Q}}\right)$ has at most two non-fixed points.*

Proof. Let

$$\pi : \mathbb{Q}[x]/(f) \to \mathbb{Q}[x]/(g)$$

be the quotient map. Then the precomposition map

$$\pi^* : \mathrm{Hom}_{\mathbb{Q}}\left(\mathbb{Q}[x]/(g), \overline{\mathbb{Q}}\right) \to \mathrm{Hom}_{\mathbb{Q}}\left(\mathbb{Q}[x]/(f), \overline{\mathbb{Q}}\right)$$

is a monomorphism of G-sets and hence the result follows from the previous lemma - 1.1. $\square$

Corollary 1.3. *Any number field K that embeds non-surjectively into $\mathbb{Q}[x]/(g)$ for any irreducible factor g of f is ramified only over primes dividing ab.*

Proof. It suffices to show that K is unramified above any prime p that does not divide ab. Let p be such a prime. It suffices to show that

$I(p)$ acts trivially on $\mathrm{Hom}_{\mathbb{Q}}\left(K, \overline{\mathbb{Q}}\right)$. Choose an embedding $\iota : K \hookrightarrow \mathbb{Q}[x] / (g)$. Then the precompositio map

$$\iota^* : \mathrm{Hom}\left(\mathbb{Q}[x] / (g), \overline{\mathbb{Q}}\right) \to \mathrm{Hom}_{\mathbb{Q}}\left(K, \overline{\mathbb{Q}}\right)$$

is a surjective G-equivariant map. Thus, $I(p)$ acts trivially on $\mathrm{Hom}_{\mathbb{Q}}\left(K, \overline{\mathbb{Q}}\right)$ since any non-trivial action of $I(p)$ on $\mathrm{Hom}_{\mathbb{Q}}\left(K, \overline{\mathbb{Q}}\right)$ would lift to an action on $\mathrm{Hom}_{\mathbb{Q}}\left(\mathbb{Q}[x] / (g), \overline{\mathbb{Q}}\right)$ that would have at least 4 non-fixed points, contradicting Corollary 1.2. □

Lemma 1.4. *A transitive permutation group that is generated by transpositions is the full symmetric group.*

Proof. Sketch inspired by a discussion with Itamar Israeli:

There is a seuqnece of transpositions that connects any two elements. Given a product of a pair of transpositions

$$(a\ b)(b\ c) = (a\ b\ c)$$

it can be shoretned to

$$(b\ c)(a\ b)(b\ c) = (a\ c)$$

$\square$

Corollary 1.5. *A primitive permutation group that contains a transposition is the full symmetric group.*

Proof. Slightly inspired by Borys's thesis:

Let H be a primitive permutation group that contains a transposition τ. Let K be the normal closure of $\langle \tau \rangle$ inside H. Then the orbits of K form a system of blocks for H. Therefore, K is itself transitive and it is generated by transpositions and hence by the previouse lemma $K = S_n$ and hence $H = S_n$ as required. $\square$

Corollary 1.6. *If the galois group G_L of a field L of degree n contains a transposition and L has no proper intermediate extensions, then $G_L = S_n$.*

Proof. According to Corollary 1.5 it suffices to show that G_L is primitive. Assume that it is not. Then there is a strictly smaller non-trivial G set S and a surjective G-equivariant map

$$\pi : \mathrm{Hom}_{\mathbb{Q}} \left(\mathbb{Q}\left(\alpha\right), \overline{\mathbb{Q}} \right) \to S$$

In particular, π is an epimorphism of G-sets, and hence by the famous equivalnece of categories between G-sets and Etale algebras over $\mathbb{Q}$, π corresponds to a monomorphism of Etale algebras over $\mathbb{Q}$, i.e. a field embedding

$$\iota : K \hookrightarrow L$$

and since S is non-trivial and strictly smaller, K is a proper subextension of $L/\mathbb{Q}$, contradicting the assumption that L has no proper intermediate extensions. $\square$

Corollary 1.7. *If g is an irreducible factor of f, there is a prime that divides the discriminant of $\mathbb{Q}[x]/(g)$ but does not divide ab, and $\mathbb{Q}[x]/(g)$ has no proper intermediate extensions, then the Galois group of g is $S_{\deg g}$.*

Proof. Let p be a prime that divides the discriminant of $\mathbb{Q}[x]/(g)$ but does not divide ab. Then by Corollary 1.2, $I(p)$ acts on $\mathrm{Hom}_{\mathbb{Q}}\left(\mathbb{Q}[x]/(g), \overline{\mathbb{Q}}\right)$ as a single transposition and hence the Galois group of g contains a transposition. The result the follows from Corollary 1.6. $\square$

Corollary 1.8. *Let g be an irreducible factor of f, and assume that for every subfield $\mathbb{Q} \subsetneq K \subseteq \mathbb{Q}[x]/(g)$ there is a prime that does not*

divide ab over which K ramifies, then the Galois group of g is $S_{\deg g}$.

Proof. According to Corollary 1.3, $\mathbb{Q}[x]/(g)$ has no non-trivial subfields, and hence, since its discriminant is divisible by a prime that does not divide a, the Galois group of g is $S_{\deg g}$ by Corollary 1.7. $\square$

Our strategy would be to find conditions on n, a, b that would guarantee the conditions of the last corollary for any irreducible factor g of f. In doing so, I found it useful to rephrase the above corollary as follows:

Corollary 1.9. *If the Galois group of L is not the full symmetric group, then there is some subextension $\mathbb{Q} \subsetneq K \subseteq L$ that is ramified only over primes dividing ab.*

1.4 $ab = \pm 1$

Assume that $ab = \pm 1$.

Corollary 1.10. *The Galois group over $\mathbb{Q}$ of any irreducible factor of f is symmetric.*

Remark 1.11. *According to Selmer* **?** *f can always be reduced to either of the forms*

$$x^n - x - 1 \quad \textit{or} \quad x^n + x + 1$$

*by substituting x by $-x$. Moreover he showed that the first form is always irreducible, while the second form is irreducible whenever $n \not\equiv 2 \mod 3$ and factors into x^2+x+1 times an irreducible factor otherwise. When $x^n \pm x+1$ is irreducible, the symmetry of its Galois group follows for example from (***?***, Theorem 1). I did not find a reference for the reducible case.*

Proof. This is immediate from Corollary 1.8 since there is no prime that divides $ab \pm 1$ and the discriminant of any non-trivial extension of $\mathbb{Q}$ is divisible by some prime as a result of the Minkowski bound. $\square$

1.5 Strategy for harder cases

Lemma 1.12. *Let p be a prime and $\mathbb{Q} \subseteq K \subseteq L$ number fields. Then*

$$v_p(\mathrm{rd}K) \leq v_p(\mathrm{rd}L)$$

and the inequality is strict iff L/K is ramified over p.

Proof. Indeed,

$$\operatorname{disc}(L) = \operatorname{disc}(K)^{[L:K]} \operatorname{Norm}_{K/\mathbb{Q}}(\operatorname{disc}(L/K))$$

and hence

$$\begin{aligned}
&v_p(\operatorname{disc}(L)) = \\
&[L:K]\, v_p(\operatorname{disc}(K)) + v_p\left(\operatorname{Norm}_{K/\mathbb{Q}}(\operatorname{disc}(L/K))\right) \geq \\
&[L:K]\, v_p(\operatorname{disc}(K))
\end{aligned}$$

and hence

$$\begin{aligned}
&v_p(\mathrm{rd}K) = \\
&\frac{v_p(\operatorname{disc}(K))}{[K:\mathbb{Q}]} \leq \\
&\frac{v_p(\operatorname{disc}(L))\,/\,[L/K]}{[K:\mathbb{Q}]} = \\
&\frac{v_p(\operatorname{disc}(L))}{[L:\mathbb{Q}]} = \\
&v_p(\mathrm{rd}L)
\end{aligned}$$

when the inequality in both of the above blocks is sharp iff L/K is ramified over p $\square$

I think the formulation of the rest of this subsection can be simplified, I will try doing it soon (In particular the constant C_p is unnecessary and I don't use it anywhere later).

Strategy 1.13. *Let p be a prime that divides ab. Using* `https://www.lmfdb.org/NumberField/` *for example, we can enumerate the first few number fields that are ramified only above primes dividing ab in ascending order of the p-valuation of their root discriminant. Thus, our strategy would be to compose a list of properties of number fields, such that any number field that satisfies one of these properties, cannot be embedded into L for any value of n (when a and b are fixed). Assume that $C_p = C_p(a,b)$ is the smallest number such that there is a number field K that is ramified only above primes dividing ab, does not satisfy any of the properties in our list, and has $v_p(\mathrm{rd}K) = C_p$. Then the Galois group of L may be non-symmetric only if $v_p(\mathrm{rd}L) \geq C_p$. Indeed, if it is non-symmetric, then by corollary 1.9, there is some subextension $\mathbb{Q} \subsetneq K \subseteq L$ that is ramified only over primes dividing ab, and hence by the definition of C_p, $v_p(\mathrm{rd}K) \geq C_p$ and hence by Lemma*

1.12, $v_p(\mathrm{rd}L) \geq v_p(\mathrm{rd}K) \geq C_p$ *as claimed.*

To compose such a list of properties, we will first prove the following:

Lemma 1.14. *Let* K *be a number field. Assume that there is an integer* $m \in \mathbb{Z}$ *and a prime* p *not dividing* ab *such that* K *has no embedding into* $\mathbb{Q}_p$ *and*

$$v_p(f(m)) = 1$$

Then K *cannot be a subfield of* $\mathbb{Q}[x]/(f)$.

Proof. Since p does not divide a, it does not ramify in K according to Corollary 1.3, and hence since K has no embedding into $\mathbb{Q}_p$, K has no embedding into any purely ramified extension of $\mathbb{Q}_p$. Therefore, to show that K can not be a subfield of $\mathbb{Q}[x]/(f)$, it suffices to show that f has a root that is contained in a purely ramified extension of $\mathbb{Q}_p$. Indeed, let e be the multiplicity of m as a root of f modulo p. Then there is a factorization

$$\pi(f) = (x-m)^e \overline{g} \in \mathbb{F}_p[x]$$

for some $\overline{g} \in \mathbb{F}_p[x]$ that is coprime to $(x-m)$ where $\pi : \mathbb{Z}_p[x] \to \mathbb{F}_p[x]$

is the quotient map. By hensel's lemma, there is a factorization

$$f = hg \in \mathbb{Z}_p [x]$$

where $\pi (h) = (x - m)^e$ and $\pi (g) = \overline{g}$. Consider now the polynomial

$$f (x + m) = h (x + m) g (x + m) = x^n + \sum_{i=0}^{n-1} c_i x^i \in \mathbb{Z} [x]$$

Then

$$v_p (c_0) = v_p (f (m)) = 1$$

and

$$\pi (f (x + m)) = \pi (h (x + m)) \pi (g (x + m)) = x^e \overline{g} (x + m) \in \mathbb{F}_p [x]$$

and hence

$$v_p (c_i) \geq 1$$

for all $i < e$. Therefore, the Newton polygon of $f (x + m)$ with respect to p has a single segment of negative and its slope is at most $1/e$ in absolute value. Therefore, $f (x + m)$ has a root, call it β, whose p-

valuation is at most $1/e$. Therefore, since the set of roots of $f(x+m)$ that reduce to 0 modulo p is exactly the set of roots of $h(x+m)$, whose degree is e, it follows that the ramification degree of β over $\mathbb{Q}_p$ is exactly its over all algebraic degree over $\mathbb{Q}_p$ and hence $\mathbb{Q}_p(\beta)/\mathbb{Q}_p$ is purely ramified. This shows that there is an embedding, in particular the one extending the assignment $x \mapsto \beta$, of $\mathbb{Q}[x]/(f)$ into a purely ramified extension of $\mathbb{Q}_p$ $\square$

Corollary 1.15. *Assume that that for every number field K that is ramified only over primes dividing ab and whose root discriminant is less than $C_p(a,b)$, either of the following holds:*

1. *K is purely imaginary and g has a real root.*

2. *There is a prime p not divide a such that $v_p(b+a+1) = 1$ and such that there is a polynomial h defining K over $\mathbb{Q}$ that has no solution mod p.*

3. *K is of even degree and there is a prime p such that $v_p(b-a+1) = 1$ and such that there is a polynomial defining K over $\mathbb{Q}$ that has no solution mod p.*

4. $v_p(\mathrm{rd}K) = v_p(\mathrm{rd}L)$ *and the ramification degree of* p *in* K *is strictly smaller than the ramification degree of* p *in* L.

Then the Galois group of g *is* $S_{\deg g}$ *if* $v_p(\mathrm{rd}L) < C_p(a,b)$ *for every prime* p *dividing* ab.

Proof. Based on Strategy 1.13, it suffices to show that any field K that satisfies one of the above conditions cannot be a subfield of $\mathbb{Q}[x]/(g)$.

1. Clear.

2. Assume that the second condition holds. Since h has no solution mod p, K has no embedding into $\mathbb{Q}_p$ and hence since

$$1 = v_p(b + a + 1) = v_p(f(1))$$

   it follows from Lemma 1.14 that K cannot be a subfield of $\mathbb{Q}[x]/(f)$.

3. Assume that the third condition holds. If n is odd, then K cannot be a subfield of $\mathbb{Q}[x]/(f)$ since the latter has an odd degree. Assume then that n is even. Again, since h has no solution mod

p, K has no embedding into $\mathbb{Q}_p$ and hence since

$$1 = v_p(b - a + 1) = v_p(f(-1))$$

it follows from Lemma 1.14 that K cannot be a subfield of $\mathbb{Q}[x]/(f)$.

4. Assume that the fourth condition holds. Then by Lemma 1.12, K cannot be a subfield of $\mathbb{Q}[x]/(f)$ since if it were, then $v_p(\mathrm{rd}K) < v_p(\mathrm{rd}L)$, contradicting the assumption of the fourth condition.

□

1.6 $2 \leq |b|$ square free and $|a| = |b|^d$ for some $d \geq 1$

Assume that $2 \leq |b|$, that b is square free and that $|a| = |b|^d$ for some $d \geq 1$. Note that in this case f is irreducible over $\mathbb{Q}$ since it is an Eisenstein polynomial with respect to the prime 2.

Lemma 1.16. *Let p be a prime dividing a. Then $v_p(\mathrm{rd}(L)) \leq d$.*

Proof. According to **?**

$$\mathrm{d}(f) = n^n b^{n-1} + (1-n)^{n-1} a^n$$

Hence, to bound $v_p\left(\mathrm{d}\left(f\right)\right)$, we will first assume towards a contradiction that

$$v_p\left(n^n b^{n-1}\right) = v_p\left(\left(1-n\right)^{n-1} a^n\right)$$

However, this equality is equivalent to

$$nv_p\left(n\right) + \left(n-1\right) v_p\left(b\right) = \left(n-1\right) v_p\left(1-n\right) + nv_p\left(a\right)$$

which holds if and only if

$$nv_p\left(n\right) - \left(n-1\right) v_p\left(n-1\right) = nv_p\left(a\right) - \left(n-1\right) v_p\left(b\right) = nd - \left(n-1\right) = \left(d-1\right) n + 1$$

Since the left hand side is divisible by p, and if $d = 1$ then the right hand side is 1, it follows that $d > 1$. Since the right hand side is positive, it follows that $nv_p\left(n\right) \geq \left(n-1\right) v_p\left(n-1\right) \geq 0$ and hence $n \equiv 0 \mod p$. However, it hence follows that the right hand side is congruent to 1 modulo p, contradicting the fact that it is divisible by p.

Therefore,

$$
\begin{aligned}
&v_p\left(\mathrm{d}\left(L\right)\right) \leq \\
&v_p\left(\mathrm{d}\left(f\right)\right) = \\
&v_p\left(n^n b^{n-1} + \left(1-n\right)^{n-1} a^n\right) = \\
&\min\left\{v_p\left(n^n b^{n-1}\right), v_p\left(\left(1-n\right)^{n-1} a^n\right)\right\} = \\
&\min\left\{n v_p\left(n\right) + \left(n-1\right), \left(n-1\right) v_p\left(1-n\right) + dn\right\} \leq dn
\end{aligned}
$$

and hence

$$v_p\left(\mathrm{rd}L\right) \leq d$$

as claimed. □

Assume now that $|a| = |b| = p$ for some prime p.

Corollary 1.17. *If there is no number field that is ramified only above p and whose root discriminant is at most p^d that does not satisfy any of the conditions in Corollary 1.15, then $Gal(f) = S_n$ for any n.*

Proof. Follows immediately from Corollary 1.15 and Lemma 1.16. □

Corollary 1.18. *The Galois group of*

$$f = x^n \pm 2x \pm 2$$

is S_n for any n.

Proof. For $n = 2$ the result is clear. Assume that $n \geq 3$. According `https://www.lmfdb.org/NumberField/?ram_quantifier=subset&ram_primes=2&rd=-2-2` the only number field that is ramified only above 2 and whose root discriminant is at most 2 is the quadratic field $\mathbb{Q}(i)$. However, the ramification degree of 2 in $\mathbb{Q}(i)$ is 2 while the ramification degree of 2 in $\mathbb{Q}[x]/(f)$ is $n > 2$ and hence it follows from Corollary 1.17 that $\mathrm{Gal}(f) = S_n$ for any n as claimed. □

Corollary 1.19. *The Galois group of*

$$f = x^n \pm 3x - 3$$

is S_n for any n.

Proof. According `https://www.lmfdb.org/NumberField/?ram_quantifier=subset&ram_primes=3&rd=-3-3` the only number field that is ramified

only above 3 and whose root discriminant is at most 3 is the quadratic field $\mathbb{Q}\left(\sqrt{-3}\right)$. Since it is purely imaginary and f has a real root, it follows from Corollary 1.17 that $\mathrm{Gal}\left(f\right) = S_n$ for any n as claimed. $\square$

Corollary 1.20. *The Galois group of*

$$f = x^n \pm 3x - 3$$

is S_n for any n.

Proof. According `https://www.lmfdb.org/NumberField/?ram_quantifier=subset&ram_primes=3&rd=-3-3` the only number field that is ramified only above 3 and whose root discriminant is at most 3 is the quadratic field $\mathbb{Q}\left(\sqrt{-3}\right)$. Since it is purely imaginary and f has a real root, it follows from Corollary 1.17 that $\mathrm{Gal}\left(f\right) = S_n$ for any n as claimed. $\square$

Corollary 1.21. *The Galois group of*

$$f = x^n \pm 4x - 2$$

is S_n for any n.

Proof. First of all note that f has a real root since $f\left(0\right) = -2$. Accord-

ing https://www.lmfdb.org/NumberField/?ram_quantifier=subset&ram_primes=2&rd=-4-4 the only number field that is ramified only above 2 whose root discriminant is at most 4 and that is not purely imaginary is $\mathbb{Q}\left(\sqrt{2}\right)$. However, it is a field of even degree, that is defined by a polynomial, $x^2 - 2$, that has no solution mod 3 while if $a = 4$ then

$$v_3\left(b + a + 1\right) = 1$$

and if $a = -4$ then

$$v_3\left(b - a + 1\right) = 1$$

Hence, it follows from Corollary 1.17 that $\mathrm{Gal}\left(f\right) = S_n$ for any n as claimed. $\square$

1.7 $a = \pm 4, b = 4$

Assume that $a = \pm 4$ and $b = 4$. The case $n = 2$ is easy - the polynomial f factors into two linear factors and hence the Galois group of any of its irreducible factors is symmetric. Assume now that $n \geq 3$.

Lemma 1.22. *f is irreducible over $\mathbb{Q}$.*

Proof. First of all, note that by analyzing the Newton polygon of f

with respect to the prime 2, it follows that all Assume towards a contradiction that f is reducible over $\mathbb{Q}$.

$\square$

1.8 $a = \pm 2, b = 1$

We will handle the case $a = -2$ first. Note that in this case 1 is a root of $f_{n,-2}$, and if we divide $f_{n,-2}$ by $x - 1$ we obtain the polynomial

$$g := x^{n-1} + x^{n-2} + \ldots + x - 1 \tag{1}$$

After substituting x by $1/x$ and multiplying by $-x^{n-1}$ we obtain the polynomial

$$x^{n-1} - x^{n-2} - \ldots - x - 1$$

which defines the same ring extension over $\mathbb{Q}$ as g and which is none other than the characteristic polynomial of the generalized Fibonacci sequence which was studied by Wolfram **?**. Wolfram showed that this polynomial is irreducible over $\mathbb{Q}$, and Martin **?** showed that whenever n is odd, its Galois group is the full symmetric group. We will show that its Galois group is the full symmetric group for any n. However,

we will stick to our version of the polynomial defined in (1).

We will denote by d the discriminant of g. Then:

Lemma 1.23.

$$v_2(d) = \begin{cases} n-2 & n \text{ is even} \\ 0 & n \text{ is odd} \end{cases}$$

Proof. By **?**

$$d = \frac{(-1)^{\binom{n}{2}} \left(n^n - 2^n (n-1)^{n-1}\right)}{(n-2)^2}$$

If n is odd it is easy to see that both the denominator and numerator are odd, and hence $v_2(d) = 0$ as required. Assume now that n is even. Then we can manipulate the expression above and get

$$d = (-1)^{\binom{n}{2}} 2^n \cdot \frac{\left(\frac{n}{2}\right)^n - (n-1)^{n-1}}{(n-2)^2}$$

and hence if we denote $m = v_2(n-2)$ we get that

$$v_2(d) = n - 2m + v_2\left(\left(\frac{n}{2}\right)^n - (n-1)^{n-1}\right)$$

Therefore, our claim is equivalent to showing that

$$n-2=n-2m+v_2\left(\left(\frac{n}{2}\right)^n-(n-1)^{n-1}\right)$$

or equivalently, that

$$v_2\left(\left(\frac{n}{2}\right)^n-(n-1)^{n-1}\right)=2m-2$$

Assume first that $m=1$, then

$$v_2\left(\frac{n}{2}\right)\geq 1$$

and hence

$$v_2\left(\left(\frac{n}{2}\right)^n\right)\geq n>0=v_2\left((n-1)^{n-1}\right)$$

and hence

$$v_2\left(\left(\frac{n}{2}\right)^n-(n-1)^{n-1}\right)=0=2m-2$$

Assume now that $m\geq 2$. In the rest of the proof we will use greek letters solely to denote different odd integer whose exact value is unknown and immaterial. As a first example for this notation, we

denote

$$n-2 \equiv \epsilon 2^m \mod 2^{2m-1}$$

From this we may deduce that

$$\begin{aligned}
&\left(\frac{n}{2}\right)^n - (n-1)^{n-1} \equiv \\
&\left(\frac{\epsilon 2^m + 2}{2}\right)^n - (\epsilon 2^m + 1)^{n-1} \equiv \\
&\left(\eta 2^{m-1} + 1\right)^n - (\epsilon 2^m + 1)^{n-1} \equiv \backslash 3m - 3 \geq 2m - 1 \\
&\binom{n}{2} \eta^2 2^{2m-2} + (n+1)\,\epsilon 2^{m-1} + 1 - (n-1)\,\epsilon 2^m - 1 \equiv \\
&\frac{(\epsilon 2^m + 2)\,(\epsilon 2^m + 1)}{2} \eta^2 2^{2m-2} + (\epsilon 2^m + 2)\,\epsilon 2^{m-1} - (\epsilon 2^m + 1)\,\epsilon 2^m \equiv \\
&\iota \eta^2 2^{2m-2} + \epsilon^2 2^{2m-1} + \epsilon 2^m - \epsilon^2 2^{2m} - \epsilon 2^m \equiv \\
&2^{2m-2}\left(\iota \eta^2 - 2\epsilon^2\right) \\
&2^{2m-2}\alpha \mod 2^{2m-1}
\end{aligned}$$

and in particular its 2-adic valuation is $2m-2$ as required. □

Corollary 1.24. *The Galois group of the non-linear factor of* $x^n - 2x + 1$ *is* S_{n-1}.

Proof. Let $\mathbb{Q} \subsetneq K \subseteq \mathbb{Q}[x]/(g)$ be a sub-extension. Then according to

Lemma 1.12

$$v_2\left(\mathrm{rd}K\right) \leq v_2\left(\mathrm{rd}\left(\mathbb{Q}\left[x\right]/\left(g\right)\right)\right) \leq \frac{v_2\left(g\right)}{n-1} \leq \frac{n-2}{n-1} < 1$$

Therefore, if $\mathrm{d}\left(K\right)$ is divisible only by 2, then $\mathrm{rd}\left(K\right) < 2$. However, theere are no number fields with root discriminant smaller than 2 that are ramified only over 2 we can prove it either with the Minkowski bound as I did in the first version of the proof, or just by citing `https://www.lmfdb.org/NumberField/?rd=-2-2` like we do in the rest of the cases . Therefore, the discriminant of K is divisible by a prime that does not divide a and hence the Galois group of g is S_{n-1} by Corollary 1.8. □

Lemma 1.25. *Assume that n is odd. Then $x^n + 2x + 1$ is irreducible over $\mathbb{Q}$.*

Proof. First of all note that $x^n + 2x + 1$ has no roots on the unit circle in $\mathbb{C}$. Indeed, if x is such a root, then

$$1 = |2x + 1| \geq |2x| - |1| = 1$$

and hence $x = -1$. However,

$$(-1)^n + 2(-1) + 1 = -1 - 2 + 1 = -2 \neq 0$$

Consider now the polynomial

$$x^n \left((1/x)^n + 2(1/x) + 1\right) = x^n + 2x^{n-1} + 1$$

whose roots are exactly the inverses of the roots of $x^n + 2x + 1$. For any small enough $\epsilon > 0$, the winding number of $x^n + 2x^{n-1} + 1$ around 0 as x traverses the circle of radius $1 + \epsilon$ is $n - 1$. Indeed, for such ϵ,

$$|2x^{n-1}| > |x^n| + 1 \geq |x^n + 1|$$

since

$$(1+\epsilon)^n - 2(1+\epsilon)^{n-1} + 1 < 0$$

since

$$\frac{d}{dx}\left(x^n - 2x^{n-1} + 1\right)|_{x=1} = n - 2(n-1) = 2 - n < 0$$

However, since $x^n + 2x + 1$ has no roots on the unit circle, and hence so does $x^n + 2x^{n-1} + 1$, it follows that all of these $n - 1$ roots are in the interior of the unit circle. In other words, only one of its roots has absolute value greater than or equal to 1 (in fact greater). If we assume towards contradiction that $x^n + 2x^{n-1} + 1$ is reducible over $\mathbb{Q}$, then since it is monic, it is reducible over $\mathbb{Z}$ as well, and hence one of its factors is a monic polynomial with integer coefficients whose roots are all of absolute value strictly less than 1 which is a contradiction (because then for example the norm, which is the free coefficient up to a sign, is less than 1 in absolute value). Therefore, $x^n + 2x^{n-1} + 1$ is irreducible over $\mathbb{Q}$ and hence so is $x^n + 2x + 1$ as required. $\square$

Corollary 1.26. *$x^n + 2x + 1$ is irreducible whenever n is odd and factors into $x + 1$ times an irreducible factor otherwise. The Galois group of any irreducible factor g of $f = x^n + 2x + 1$ is $S_{\deg g}$.*

Proof. Assume first that n is even, then

$$(-x)^n + 2(-x) + 1 = x^n - 2x + 1$$

and hence by the above analysis, $x^n - 2x + 1$ factors into $x + 1$ times

an irreducible factor whose Galois group is S_{n-1}.

Assume now that n is odd. According to (**?**, Theorem 2)

$$\mathrm{d}\left(f_{n,2}\right) = (-1)^{\frac{n(n-1)}{2}} \left(n^n + (1-n)^{n-1} 2^n\right)$$

and hence $v_2\left(\mathrm{d}\left(f_{n,2}\right)\right) = 0$. Therefore, discriminant of any intermediate extension $\mathbb{Q} \subsetneq K \subseteq \mathbb{Q}[x] / (f_{n,2})$ is, odd and in particular it is divisible by a prime that does not divide 2 and hence the Galois group of $f_{n,2}$ is S_n by Corollary 1.8. $\square$

1.9 $a = \pm 3, \pm 4, \pm 5, b = 1$

Lemma 1.27. *If $|a| > 2$ then $f_{n,a}$ is irreducible over $\mathbb{Q}$.*

Proof. It suffices to show that

$$\widehat{f_{n,a}} := x^n f_{n,a}(1/x) = x^n + ax^{n-1} + 1$$

is irreducible over $\mathbb{Q}$. For any $x \in S^1 \subseteq \mathbb{C}$,

$$\left|ax^{n-1}\right| = |a| > 2 = |x^n| + |1| \geq |x^n + 1|$$

and hence the winding number of $\widehat{f_{n,a}}(x)$ around 0 as x traverses the unit circle is equal to that of $-ax^{n-1}$ which is $n-1$. Therefore, $\widehat{f_{n,a}}$ has $n-1$ roots in the interior of the unit disk and hence it is irreducible over $\mathbb{Q}$. $\square$

Remark 1.28. *The above lemma also follows from (**?**, Theorem 1).*

Lemma 1.29. *Let p be a prime dividing a. If*

$$v_p(n) \neq v_p(a) \;\; or \;\; (n/a)^n \not\equiv -1 \mod p$$

then

$$v_p(disc(f_{n,a})) = n \min\{v_p(n), v_p(a)\}$$

Proof. Assume first that $v_p(n) \neq v_p(a)$. According to (**?**, Theorem 2)

$$\operatorname{disc}(f_{n,a}) = (-1)^{\frac{n(n-1)}{2}} \left(n^n + (1-n)^{n-1} a^n\right)$$

Then either $v_p(n) = 0$ and hence $v_p(\operatorname{disc}(f_{n,a})) = 0$ or $v_p(n) > 0$ and hence $v_p(n-1) = 0$ and thus $v_p\left((1-n)^{n-1} a^n\right) = nv_p(a)$ and hence

$$v_p(\operatorname{disc}(f_{n,a})) = \min\left\{v_p(n^n), v_p\left((1-n)^{n-1} a^n\right)\right\} = n \min\{v_p(n), v_p(a)\}$$

Assume now that $v_p(n) = v_p(a)$. Then

$$\operatorname{disc}(f_{n,a}) = \operatorname{Res}\left(f_{n,a}, f'_{n,a}\right) = p^{nv_p(a)}\operatorname{Res}\left(f_{n,a}, p^{-v_p(a)} f'_{n,a}\right)$$

Since $p^{-v_p(a)} f'_{n,a} \in \mathbb{Z}[x]$ it follows that $v_p(\operatorname{disc}(f_{n,a})) \geq nv_p(a)$ and equality holds iff $f_{n,a}$ and $p^{-v_p(a)} f'_{n,a}$ have no common root in $\overline{\mathbb{F}_p}$. Assume then that there is no equality, i.e. that

$$\begin{aligned}
\overline{f_{n,a}} &= x^n + ax + 1 = x^n + 1 \in \mathbb{F}_p[x] \\
p^{-v_p(a)}\overline{f_{n,a}}' &= np^{-v_p(a)}x^{n-1} + ap^{-v_p(a)} \in \mathbb{F}_p[x]
\end{aligned}$$

have a common root. This root in particulat annahilates

$$g_{n,a} := x\overline{f_{n,a}}' - n\overline{f_{n,a}} = nx^n + ax - nx^n - n = ax - n$$

i.e. it is equal to n/a and hence

$$0 = \overline{f_{n,a}}(n/a) = (n/a)^n + 1$$

and thus $(n/a)^n \equiv -1 \mod p$. $\square$

Let $S(|a|)$ be the set of positive integers n that satisfy the conditions of Lemma 1.29, i.e. for any prime $p \mid a$ either

$$v_p(n) \neq v_p(a) \text{ or } (n/a)^n \not\equiv -1 \mod p$$

Corollary 1.30. *Let $n \in S(|a|)$. Then*

$$v_p(\mathrm{rd}(f)) \leq v_p(a)$$

Assume now that a is a power of some prime p.

Corollary 1.31. *If any number field that is ramified only over p and whose root discriminant is at most a in absolute value satisfies one of the conditions of corollary 1.15 then for any $n \in S(a)$, $Gal(f) = S_n$.*

Corollary 1.32. *For any $n \in S(3)$, $Gal(f_{n,\pm 3}) = S_n$.*

Proof. According to `https://www.lmfdb.org/NumberField/?ram_quantifier=subset&ram_primes=3&rd=-3-3`, the only non-trvial number field that is ramified only over 3 and whose root discriminant is at most 3 is the cyclotomic extension $\mathbb{Q}(\zeta_3) = \mathbb{Q}\left(\sqrt{-3}\right)$ which is purely imaginary and hence the result follows from Corollary 1.31. $\square$

Corollary 1.33. *For any* $n \in S(4)$, $Gal(f_{n,\pm 4}) = S_n$.

Proof. According to `https://www.lmfdb.org/NumberField/?ram_quantifier=subset&ram_primes=2&rd=-4-4&sort_order=rd&sort_order=rd`, the only non-trivial number fields that are ramified only over 2 and whose root discriminant is at most 4 are the three purely imaginary fields defined by the polynomials x^2+1, x^2+2, x^4+1, and the quadratic field $\mathbb{Q}\left(\sqrt{2}\right)$ which is defined by x^2-2 which has no solution mod 3 which divides $4+2=-(-4)+2$, is inert, and hence the result follows from Corollary 1.31. □

Corollary 1.34. *For any* $n \in S(5)$, $Gal(f_{n,\pm 5}) = S_n$.

Proof. According to `https://www.lmfdb.org/NumberField/?ram_quantifier=subset&ram_primes=5&rd=-5-5&sort_order=rd&sort_order=rd`, the only non-trivial number fields that are ramified only over 5 and whose root discriminant is at most 5 are the cyclotomic extension $\mathbb{Q}(\zeta_5)$ and the quadratic field $\mathbb{Q}\left(\sqrt{5}\right)$. The first is purely imaginary, and the second may be defined by x^2-5 which has no solution either mod 7 which divides $5+2$ or mod 3 which divides $-5+2$ and hence the result follows from Corollary 1.31. □

I cannot reliably check any largegr values since the database is guaranteed to be complete only for root discriminants up to 5.989.

The following table summerizes the set $S(a)$ for each $a = \pm 3, \pm 4, \pm 5$:

a	$n \in S(a) \Leftrightarrow n \not\equiv$	mod
2	2	4
± 3	$-a$	18
± 4	4	8
5	$10, 40, 45, 60, 90, 95$	100
-5	$5, 10, 40, 55, 60, 90$	100

Remark 1.35. *The cases where n and a are coprime, follow from (**?**, Theorem 1). For the other cases I did not find any published proof.*